Divyesh K

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SUMMARY

Motivated AI Developer having hands-on experience in Python, ML, Neural Networks, Quantization and AI pipelines. Developed Quantized models for Nucleotide Transformers, ML Pipelines, Fault detection engines for smart grids, Firmware level ML based PQC managers. Actively seeking opportunities in AI and ML roles.

EDUCATION

Bachelor of Engineering in Electronics and Communication Engineering

SRM Institute of Science and Technology (Ramapuram) • Tamil Nadu, India • 2023-2027 • CGPA: 9.21/10

EXPERIENCE

Summer Intern

National Institute of Technology (NIT) Trichy **June 2025 - July 2025, Trichy, TN**

- Developed a custom fault detection engine by exploring DLPF solvers, ML frameworks, and power system behaviour for CIGRE-MV networks detection and localizing single bus faults.
- Generated 21,000+ power grid scenarios (faults, load variations) using pandapower which increased accuracy to 95%.
- Implemented ensemble methods comprising multi-tier random forest trees achieving 92% detection and 100% localization of grid faults.
- Improved solver using DLPF, reducing voltage computation time by 70% vs. Newton–Raphson.

MERN Developer Intern

Circle Communications **May 2025, Chennai, TN**

- Engineered high-performance full-stack CRM with Redux-powered frontend (59k+ LOC modular UI) and microservice backend (1.2k SLOC), delivering 51ms API latency for real-time enterprise interactions.
- Applied "Thin Server, Thick Client" architecture, offloading complexity to 300k+ dependency libraries while maintaining lean business logic for scalability and zero technical debt.
- Enhanced DX using Vite + Concurrently, slashing full-stack cold startup to 1.39s—maximizing developer velocity in agile environments.
- Built resilient backend with auto-retry DB connections, ensuring 99.9% uptime against infrastructure failures in production simulations.

PROJECT

SleeQC: Resource-Adaptive Post-Quantum Cryptography on ESP32

SRM University • github.com/Divyesh-Kamalanaban/sleeqc • August 2025 - October 2025

- Pioneered a Resource-Adaptive Post-Quantum Cryptography (PQC) system on an edge-constrained ESP32-S3 to dynamically balance high security
  and real-time operational stability.
- Deployed a TinyML decision engine (TensorFlow Lite) that monitors real-time system load (Free Heap, task duration) to autonomously switch between ML-DSA-44 (Dilithium2) and ML-DSA-87 (Dilithium5).
- Optimized system stability for resource-intensive cryptography by triaging and resolving critical stack overflow errors, increasing the worker task stack size to 16KB to ensure stable execution of Dilithium5.
- Streamlined cross-language integration within the ESP-IDF framework by resolving complex C/C++ linker issues and managing custom component dependencies.

Identifying Taxonomy and Assessing Biodiversity from eDNA Datasets

SIH 2025 • github.com/Divyesh-Kamalanaban/seaquenced • August 2025 - October 2025

- Designed core components of the pipeline to classify species from environmental DNA (eDNA) datasets.
- Expanded dataset from 5 species FASTQ files by mutating sequences, generating 10,000+ training samples.
- Quantized Nucleotide-Transformer from 32-bit to 4-bit precision, reducing GPU memory usage from 9.5GB to 2.5GB with negligible accuracy loss.
- Contributed and led multiple stages of the project, supporting model training, integration, and preprocessing workflows.

SKILLS & CERTIFICATIONS

Languages, Frameworks & Tools: Python, JavaScript, React.js, Node.js, Express, MongoDB, Tailwind CSS, Git, TensorFlow

Libraries & Technologies: NumPy, Pandas, REST APIs, Deep Neural Networks (DNN), Random Forests, scikit-learn, Keras, TensorFlow Lite Micro, Linux, XGBoost, PyTorch, Transformers, LangChain, LangGraph, Ollama

Certifications: GitHub Foundation